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| **PI Name:** | Dunn-Rankin, Derek Ph.D. | | |
| **Project Title:** | Applications of Electric Fields in Microgravity Combustion Science | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | COMBUSTION SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles | | |
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| **Zip Code:** | 92697-3975 | **Congressional District:** | 48 |
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| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Weinberg, Felix ( Imperial College of Science, Technology, and Medicine ) | | |
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| **Performance Goal Text:** | | | |
| | Transport control by large electric fields in microgravity has a variety of potential applications that include pumps with no moving parts, enhanced particle collection devices, application of liquid films to surfaces in the absence of gravity, and control of flames when natural convection is absent. Using electric fields to induce and direct convection through the drag exercised on neutral gas by ions and to manipulate dispersions and trajectories of electrically charged droplets and particles is especially effective in the absence of earth gravity. Transport control by large electric fields offers numerous applications which may be subdivided according to whether the charge carriers are flame ions, ions produced by corona discharges, or electrically charged droplets and particles. Flame ions and ions produced by corona discharges have similar attributes, except that the maximum current densities due to the former are limited by their rate of generation by chemi-ionization in the reaction zone, whereas the supply of the latter continues to increase approximately | | |

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| **Task Description:** | quadratically with the applied potential until breakdown. Droplets and particles are deliberately charged during their dispersion process in our studies and manifest very much smaller mobilities but the forces that can be applied to them electrically relative to those of gravity, centrifugal acceleration, and convective drag can be substantial. In order to achieve the goal of this project, which is to understand the opportunities for controlling transport in microgravity, there are three principal thrusts to the research:<br>(1) the electrically controlled flame system, where experiments are conducted in an Earth based laboratory enabling a wide range of non-intrusive diagnostics on flames in various g-loading conditions including that equivalent to microgravity. Diagnostics such as schlieren, shadowgraph, deflection imaging, are applied to visualize a burning methane-air flame under the influence of electrically induced microbuoyancy. Numerical simulations predict the forces involved and allow new designs for enhanced flame control. (2) ion driven winds for ventilation, electrostatic precipitation, and burners. For these applications it is necessary to understand the limitations of ion winds. For example, it is to be determined whether or not the airflow can be accelerated with the addition of a nozzle and/or if the blowers are staged. From the results, an ionic air swirler is to be designed where a vortex is formed within a tube containing air at STP with adequate strength to hold a flame. The vortical flow can also be applied to a fan-less electrostatic precipitator capable of enhanced particle removal efficiency by combining electrostatic and centrifugal forces on the particles entrained into the system. (3) electrosprays, where electrically charged dispersions of liquids or solid particles are used to extinguish fires or as surface coatings for therapeutic agents.<br><br>Understanding electric field interactions with flames is central to combustion control, a technology that not has application in advanced propulsion and power systems, but also toward fire safety in space and on other planets. Appropriately applied electric fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our facility provides a unique ability to make simultaneous measurements of temperature, species concentration, flow patterns, and ion currents.<br><br>One goal central to NASA's mission is to improve spacecraft and remote habitat fire safety for travel and off-world activities. The ion wind and electrospray work is expected to reveal new and improved ways to prevent, detect, and extinguish fires. In addition, advances in small-scale combustion systems capable of providing compact power can be developed without moving parts. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Several of the applications we explore have potential uses in earth gravity as well as microgravity. For example, the application of liquid dispersions of biologically active material for agriculture can reduce the use of chemical toxins. Understanding fundamental electrical properties of flames will also have contributions to earth-based combustion though the forces from electric fields will be less dramatic in the presence of natural convection. |
| | Taking the above three project foci in turn, the following progress has been achieved over the past year of the project -- Electrially controlled flames – The experimental apparatus we use in our study allows full optical access to allow schlieren, chemiluminescence, and other spectroscopic measurements. In addition, we can measure the total ion current from the flame and the distribution of ion flux at the ground electrode. This information combines to provide a physical description of the body force balance between electric winds and buoyancy as well as quantitative date that can be compared to numerical simulations of the capillary flame. The schlieren images indicate clearly that the point of balance is not a global condition, but a local one as there is a distinct plume of gas visible even when the flame is nearly spherical.<br><br>In order to understand this region of balance more completely, the distribution of ion flux (which is linearly related to the electric body force – when the ion mobility is constant). We have found that the distribution of ions from a flame and from a point corona source are identical, and this is significant because it suggests that the ion distribution in the near-flame region is governed by self-repulsion of ions and the resulting space charge more than it depends on the electric field set up by the external electrodes. The disappointing corollary to this finding is that it will be difficult to control the extent of the force-balance condition by manipulating the electrode geometry. The electric force (governed by the distribution of ions) does not match inversely the buoyant force for an ideal spherical temperature gradient. Because these two functions do not match, the region of balance extends only over a limited region (approximately 20 degrees). This value agrees with our prior measurement results.<br><br>In order to examine the feasibility of this interpretation, we have constructed a comprehensive CFD model of the capillary flame in an electric field using FEMLAB. So far, we have simulated the flame as a constant potential surface that is also a source of ions. We then predict the path of the ions and compare calculated ion currents to experiments. The simulations show that there is a limited region over which the ions can extend before they are dragged to the ground electrode. This region coincides very closely to the region we have found through experiment to be the location of a shear layer separating an outer buoyant plume and the fairly stagnant inner balance region. Good agreement has been obtained between experimental and computed ion profiles. We then included the fluid flow coupling in the model. The flow pattern computed is similar to that seen in schlieren images. Interestingly, a recirculation pattern occurs whether the ground plane is porous or solid, explaining to some extent the unstable nature of the flame balance condition. We have also computed the response of the flame to a step change in the electric field in order to understand the time delay between the onset of electric force (which is essentially instantaneous) and the change in flame shape (which takes nearly 100 ms). The computation indicates that the time is taken to set up a vortex. This vortex establishes the pressure field and flow distribution that ultimately controls the flame shape and behavior. It is possible to take these calculations further by adding a chemical kinetic mechanism, but we will first examine the role of electrode geometry to see if we can create a balanced force environment that extends further than our existing system creates.<br><br>Ion Driven Winds – We have completed a comprehensive study of how to create and optimize an ion driven wind in the configuration where a high potential is maintained between a needle electrode and a ring electrode in a tube. The highly charged needle produces ions that are swept toward the ring by the electric field. These ions then produce a wind by dragging along the neutral air molecules. We have found several geometrical sensitivities in this system, including the distance of the needle to the ring, the width of the ring, the accuracy with which the needle is centered, and the overall obstructions to the flow. We have also discovered that staging needle-tube ion wind generators (i.e., stacking them in series) does not produce ever-higher flow rates. There is a limiting flowrate, which indicates that the ion wind process creates an aerodynamic loss, much as does a pipe corner or valve. We have further found that it is possible to increase |

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| **Task Progress:** | the flow velocity of the ion wind by stacking generators in series and then putting a nozzle at the exit. A nozzle at the end of the series of stages can then create sufficient pressure to drive acceleration through the nozzle while keeping the flow through the ion wind stages low enough that the losses do not remove the advantages. We have studied from 1-10 stages and from this data we have created a complete fan map of the ion driven wind. The “fan curve” is a comprehensive description of pin-tube ion driven wind generation. It shows that each stage produces approximately 5.5 Pa of pressure increase. The data from this figure, combined with voltage and current measurements, provides fan efficiency information in terms of the ratio of flow power to electrical power used. These ion-driven wind devices have efficiency in the range of 0.1%, which while low, may be acceptable for small scale systems where traditional fans cannot operate. This electrical efficiency is a factor of 10 lower in corona ion winds versus flame ion winds because most of the loss in the former occurs during the electron cascade producing the ions.<br><br>In order to optimize the performance of pin-tube ion wind generation, we have created a FEMLAB simulation of the system (just as in the flame case). The simulation assumes a fixed ion concentration at the needle to act as the source and matches the current flowing through the system to produce a quantitative match. The domain of the calculation extends far from the flow tube in the inlet direction to accurately predict the pressure drop that occurs as the ion pump drives flow through the system. We have also compared the velocity profile measured using a small pitot tube and the velocity profile calculated. There are some substantial quantitative differences, but the integrated flux and the profile behavior are well predicted. The challenge for the calculation is that it assumes a laminar flow streaming off of the needle tip. In reality, the corona discharge is not smooth and steady but fluctuating both in position and time. This gives rise to a broader velocity profile near the needle rather than the impossibly narrow jet that appears in the laminar prediction. In addition, as with all jets, the axial flow from the needle tip would rapidly entrain the slower moving air nearby, but without a good turbulence model, this entrainment is underpredicted. We have simulated its effect to some extent by artificially increasing the diffusion. We have also attempted to include a standard turbulence model, but because the calculation requires very fine gridding to capture the high electrical gradients, we have insufficient computer resources to fully model the system. Although disappointing, for our purposes, the artificial diffusion model is sufficient because it allows us to examine geometric effects for optimal performance, which was the primary goal of the effort. We are in the process of computing total flux as a function of needle-to-ring position, length of the ring electrode, and tube diameter. This information should allow a complete system performance prediction for ion wind generators of this design.<br><br>Using ion winds to create swirling flows – We have studied the use of a swirling flow generated by ion winds for improving the performance of electrostatic precipitation, but our principal interest is using swirl to improve burners. Based on our swirling flow results, we know how to best create a cyclone. Our intent is to use a wicked fuel at the center of a swirling flow. To simulate this system we started with a gaseous flame.<br><br>Evaluate the use of hand-held electrosprays – We will be studying the use of electrosprays to extinguish fires in a microgravity environment, but we have also identified an exciting new use of electrosprays that may be relevant to space and space exploration. In particular, we are looking at electrosprays of biologically active materials to see if they remain viable after processing. Biologically viable material can be used as more effective nutrient and pesticide than chemicals in agriculture. It is also likely that this material will have use for topical medical purposes.<br><br>Further information on all of the progress and activities described above can be found in the papers listed in the references section of this report. |

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| **Bibliography Type:** | Description: (Last Updated: 11/26/2025) |
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| **Dissertations and Theses** | Rickard, M.A. "Corona Ion Driven Winds" Ph.D. Dissertation, UC Irvine<br>Oct-2005 |
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